KOTHAMASU MOHAN VENKATA SUBBA RAO

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PROFESSIONAL SUMMARY

Final-year B.Tech Information Technology student (graduating in 2027) with a solid foundation in Python, SQL, Java, and Web Development. Experienced in building practical software applications, IoT telemetry hardware prototypes, and algorithmic problem-solving. Seeking an entry-level Software Engineer, Python Developer, or Frontend Developer role to build reliable applications, write clean modular code, and solve business challenges.

TECHNICAL SKILLS

Programming Languages: Python, Java, C (Basics)
Web Development: HTML5, CSS3, JavaScript (DOM Manipulation, Responsive Design)
Databases: SQL, MySQL (Schema Design, Complex Joins, Aggregations, Subqueries)
Core Concepts: Computer Networks (TCP/IP, OSI Layers), Object-Oriented Programming (OOP), Data Structures
Tools & Platforms: Git, GitHub, VS Code, Arduino IDE, Microcontrollers (ESP32, Arduino)

EDUCATION

QIS College of Engineering and Technology — Ongole, India | Expected 2027
Bachelor of Technology (B.Tech) in Information Technology — ***CGPA: 7.50 / 10.0***

Sri Chaitanya Junior College — Addanki, India | 2021 – 2023
Intermediate (Class 12 – MPC) — *Score: 821 / 1000 (82.1%)*

Holy Faith School — Addanki, India | 2020 – 2021
Secondary School Certificate (SSC – Class 10) — *Score: 600 / 600 (100% – Perfect Score)*

TECHNICAL PROJECTS

Smart Autonomous Irrigation System — Embedded C/C++, Microcontroller, Automation
Four-bar mechanism valve control with water-level feedback and safety timer — Live Demo → GitHub →

- Engineered an autonomous irrigation control system automating pump activation based on real-time soil moisture and water-level sensing.
- Implemented hardware fail-safe timer fallback logic to prevent pump dry-running, safeguard motor longevity, and eliminate agricultural water waste.
- Integrated a four-bar linkage mechanical assembly for targeted and uniform water delivery across defined plantation zones.

IoT Air Quality & Hazardous Gas Monitoring System — ESP32, MQ Sensors (MQ2/MQ7/MQ135), DHT11, IoT
Atmospheric multi-sensor telemetry with real-time threshold alert triggers — Project Docs → Live Demo → GitHub →

- Built a multi-sensor atmospheric telemetry unit using ESP32 and MQ sensor arrays to detect toxic gases, carbon monoxide, and smoke.
- Programmed real-time threshold alert triggers and integrated a DHT11 sensor for continuous ambient temperature and humidity tracking.
- Structured lightweight data telemetry for environmental safety monitoring in hospital and public facility enclosures.

Piezoelectric Power Generation Footwear — Arduino UNO, Piezoelectric Sensors, Energy Harvesting
Kinetic footstep pressure conversion into regulated electrical power — Prototype Code → Live Demo → GitHub →

- Designed a kinetic energy harvesting prototype converting footstep pressure into regulated electrical charge stored in capacitor banks.
- Integrated an LCD telemetry display with Arduino UNO to visualize real-time generated voltage metrics and power efficiency.

EXPERIENCE & INTERNSHIPS

Python Full Stack Virtual Internship — AICTE – EduSkills / Edunet
Software & Web Development Intern — Verify Certificate →

- Completed intensive hands-on training on Python programming, object-oriented concepts, and relational database management.
- Built modular full-stack web components using HTML5, CSS3, JavaScript, and backend Python scripts with SQL database integration.
- Practiced version control workflows with Git and GitHub, and executed database query optimization exercises.

CERTIFICATIONS & ACHIEVEMENTS

- **NPTEL Elite Certification:** Developing Soft Skills and Personality / IoT – Scored **78% (Elite)** View Certificate →
- **AWS Academy – AICTE / EduSkills:** AI-ML Virtual Internship (10 Weeks) – Scored **Grade A (Very Good)** View Certificate →
- **Python Full Stack Certification:** AICTE / Edunet Foundation View Certificate →
- **Tata GenAI / Data Analytics Job Simulation:** Industry workflows on Forage Verify Certificate →
- **All-India Rank 6000:** AINCAT Nationwide Competitive Technical Assessment Scorecard →